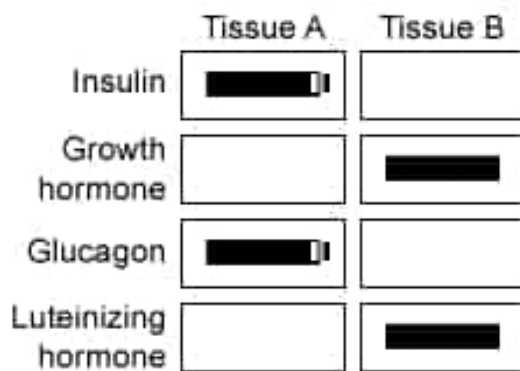


The diagram below shows the results of a northern blot analysis that tested for the expression of four peptide hormones in two endocrine tissues.



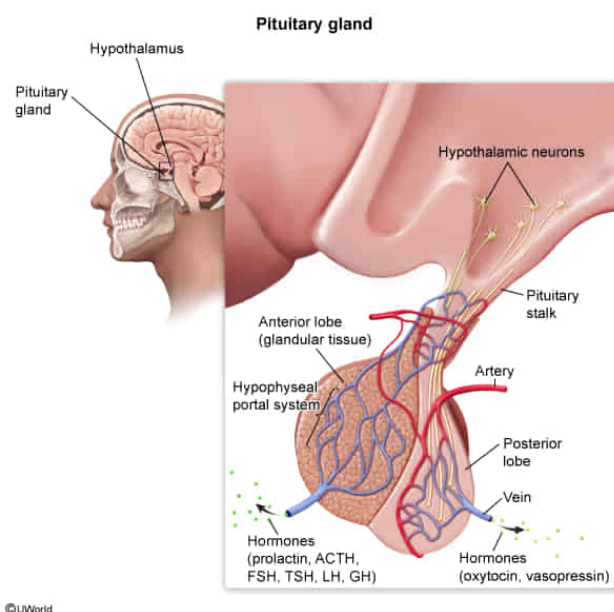
The endocrine glands that most likely correspond to tissues A and B are the:

- ☐ A. pituitary gland and pancreas, respectively.
- ✓ ☒ B. pancreas and pituitary gland, respectively. [95%]
- ☐ C. adrenal glands and hypothalamus, respectively. [2%]
- ☐ D. hypothalamus and adrenal glands, respectively.

Correct

95% answered correctly

Explanation:



The **endocrine system** modulates physiological activity via the secretion of hormones, which **travel** through the bloodstream and bind receptors in target tissues to elicit specific responses. The following are examples of endocrine glands:

- The anterior lobe of the pituitary gland secretes several hormones, including thyroid-stimulating hormone (TSH), luteinizing hormone (LH), follicle-stimulating hormone (FSH), adrenocorticotrophic hormone (ACTH), and growth hormone (GH). Specifically, GH promotes processes such as tissue growth and utilization of fats. In addition, pituitary release of LH and FSH regulates reproductive function in males and females. In females, FSH and LH stimulate follicle maturation, ovulation, and estrogen production. In males, **spermatogenesis** and testosterone production are regulated by FSH and LH, respectively.
- The pancreas regulates **blood glucose concentration** by secreting **insulin and glucagon**. Insulin is secreted in response to high blood glucose levels to promote glycogen formation in the liver and uptake of glucose by cells. In contrast, glucagon is secreted in response to low glucose levels to promote gluconeogenesis and glycogen breakdown.

In the given scenario, **northern blotting** was used to test mRNA expression in two endocrine tissues. The **results** show that tissue A (the pancreas) expresses insulin and glucagon, and tissue B (the anterior pituitary) expresses GH and LH.

(Choice A) This scenario is reversed; the pancreas does *not* secrete LH and GH, and the pituitary gland does *not* secrete insulin and glucagon.

(Choices C and D) The adrenal glands secrete hormones that modulate the physiological **stress response**, and the hypothalamus secretes several hormones (eg, GRH, GnRH) that *regulate* release of pituitary hormones. However, neither of these endocrine glands secrete the hormones shown in the northern blot.

Educational objective:

Endocrine glands modulate physiological activity via the secretion of hormones. For example, the pancreas secretes insulin and glucagon to regulate blood glucose levels. In contrast, the anterior lobe of the pituitary gland secretes multiple hormones that regulate several processes, including metabolism and reproductive function.

Time spent:0 sec

Subject:
Biology

QID:402213

System:
Endocrine and Nervous Systems

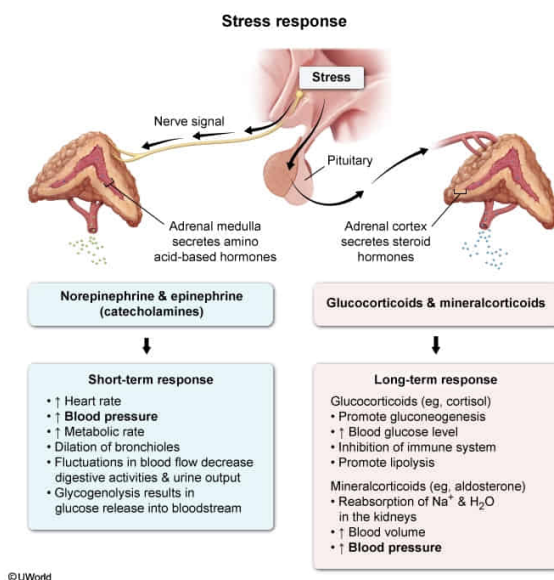
As the two anatomical divisions of the adrenal gland, the adrenal cortex and adrenal medulla share a common feature in that both:

- ☐ A. synthesize steroid hormones. [21%]
- ☐ B. release signaling molecules into lymph vessels. [9%]
- ☐ C. regulate the synthesis of red blood cells. [1%]
- ✓ ☒ D. secrete hormones that alter blood pressure. [68%]

Correct

67% answered correctly

Explanation:



The adrenal glands are **endocrine organs** located on top of the kidneys that secrete hormones mediating the stress response (physiological changes that allow the body to cope with and react to physical, social, and emotional stressors). The **adrenal cortex** secretes the hormones **cortisol** and **aldosterone**, and the **adrenal medulla** secretes the hormones **epinephrine** and **norepinephrine**.

Aldosterone, secreted from the adrenal cortex, stimulates **Na^+ reabsorption** in the kidneys, *leading to increased water retention* due to osmosis. Increased water retention *increases blood volume*, which in turn leads to an *increase in blood pressure*.

Norepinephrine and *epinephrine*, secreted from the adrenal medulla, function to mobilize the body under extreme stress and promote rapid information processing in part by maximizing blood flow to organs essential for survival. Norepinephrine and epinephrine achieve this effect by promoting the following changes in **blood vessel diameter** that lead to altered blood pressure:

- **Vasoconstriction** (narrowing) of blood vessels supplying the intestines, kidney, and other abdominal organs decreases blood flow to these organs and conserves

oxygen/nutrients for other tissues.

- *Vasodilation* (widening) of blood vessels leading to the heart and skeletal muscles increases oxygen/nutrient delivery to these organs.

In general, the **hormones secreted from both the adrenal medulla and cortex function to alter blood pressure.**

(Choice A) The adrenal cortex synthesizes and secretes *lipophilic hormones* derived from *cholesterol*, otherwise known as **steroid hormones**. However, the adrenal medulla secretes the *peptide* hormones norepinephrine and epinephrine, which are derived from amino acids (*not* cholesterol).

(Choice B) Endocrine glands such as the adrenal gland secrete hormones into the blood vessels, *not* lymph vessels. Adrenal hormones are distributed throughout the body via the circulatory system.

(Choice C) In response to low oxygen levels, the hormone erythropoietin is released from the kidneys (*not* the adrenal glands) and acts on the bone marrow to stimulate red blood cell production.

Educational objective:

Each adrenal gland is composed of two anatomically distinct regions: the adrenal medulla and the adrenal cortex. Both regions secrete hormones that regulate blood pressure and allow the body to respond to stressors.

Time spent:0 sec

Subject:

Biology

QID:402214

System:

Endocrine and Nervous Systems

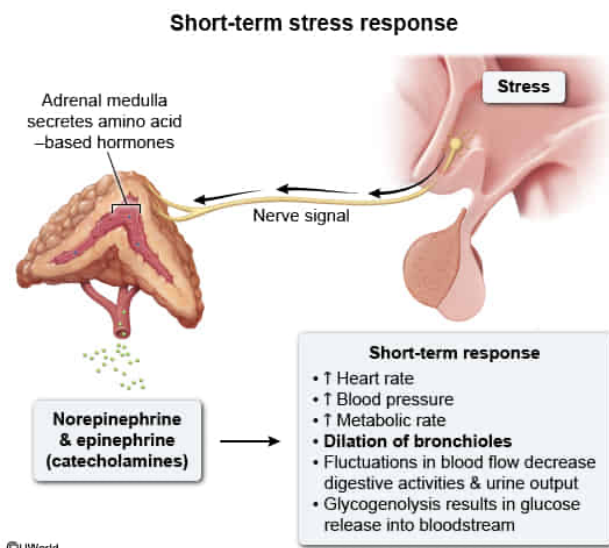
One function of catecholamines released by the adrenal glands during times of stress is to:

- ☐ A. stimulate secretion of melatonin by the pineal gland. [1%]
- ✓ ☒ B. increase dilation of bronchioles in the lungs. [77%]
- ☐ C. inhibit glucose reabsorption by the kidneys. [18%]
- ☐ D. promote the release of digestive enzymes. [2%]

Correct

76% answered correctly

Explanation:



The adrenal glands are divided into two subdivisions, the adrenal cortex and adrenal medulla. Hormones secreted by the adrenal medulla and cortex modulate function in a way that enables the body to respond to stressful situations. Specifically, adrenal hormones promote the "fight-or-flight" response.

For example, in **response to a stressor**, the **adrenal medulla** releases **catecholamines** (norepinephrine and epinephrine) that induce the following physiological changes:

- Redirection of blood flow throughout the body via **vasoconstriction and vasodilation**
 - Vasoconstriction (narrowing) of blood vessels and reduced blood supply to organs that carry out nonessential functions (eg, stomach, intestines, kidneys) conserve oxygen and nutrients for organs that are necessary for immediate survival.

- Vasodilation (widening) of blood vessels increases blood flow to organs essential for immediate survival (eg, heart, skeletal muscle).
- Increased heart rate and cardiac muscle contractility promote increased blood flow to the brain, lungs, and skeletal muscles, allowing an organism to process and execute a response to the stressful stimulus.
- **Dilation of airways** (bronchioles) enables increased respiratory function and oxygen delivery to tissues.

(Choice A) Melatonin, a hormone that promotes drowsiness and sleep, is released by the pineal gland in response to **low light levels** (detected by the retina). Adrenal gland activity is *not* involved in the regulation of the pineal gland.

(Choice C) Although glucose reabsorption occurs in the kidney, adrenal hormones (eg, aldosterone) do not affect this process. However, aldosterone *does* stimulate the **reabsorption** of sodium and water in the kidneys.

(Choice D) By decreasing blood flow to the stomach and other digestive organs, the actions of hormones secreted by the adrenal gland will *decrease* digestive activity, likely *inhibiting* the secretion of the digestive enzymes that break down food.

Educational objective:

During the stress response, catecholamines secreted from the adrenal medulla act to promote the "fight-or-flight" response. Some effects of catecholamines include redirected blood flow to maximize the delivery of oxygen and other nutrients to organs essential for immediate survival (eg, brain, lungs, skeletal muscles), increased heart rate, and dilated airways.

Time spent:0 sec
Subject:
Biology

QID:402215
System:
Endocrine and Nervous Systems